LINEAR ALGEBRA · ONE-PAGE REFERENCE

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Scope: MIT 18.06SC, first-half topic selection through determinants and volume; not an official exam scope. No eigenvalues, diagonalization or SVD.
Real $A \in \mathbb{R}^{m\times n}$; $r = \operatorname{rank}(A)$; $R = \operatorname{rref}(A)$; $C(A)$=column space; $N(A)$=nullspace; T=transpose. Black: explanations / conditions; blue: formulas; red: traps.

01 Systems: three pictures [S01 p1–3; S02 p1–4]

Rows: intersect the equation hyperplanes.
Columns: combine the columns a_j to obtain b.
Matrix: write both pictures as one equation.

$$Ax = \sum_{j=1}^{n} x_j a_j = b$$

Solving a system means finding the coefficients of that linear combination. $a_j, b \in \mathbb{R}^m$; $x \in \mathbb{R}^n$.

02 Products and transposes [S04 p1–3; S05 p1]

$A_{m\times n}B_{n\times k}$ has size $m \times k$. The inner dimensions must agree.

$$(AB)_{ij} = \sum_{\ell=1}^{n} a_{i\ell}b_{\ell j}$$

Columns of AB combine columns of A; rows of AB combine rows of B. Outer-product form:

$$AB = \sum_{\ell=1}^{n} A_{:\ell}B_{\ell:}$$

$$(AB)^T = B^T A^T, \quad (AB)^{-1} = B^{-1}A^{-1}$$

The inverse rule requires invertible square factors.
Watch: Order reverses. Generally $AB \neq BA$; A^TA is always symmetric.

03 Elimination, permutations, LU [S03 p1–3; S05 p1–3; S06 p1]

Apply $R_i \leftarrow R_i - \ell_{ij}R_j$ to both A and b. Swap rows for a zero pivot; a rectangular system may have free columns.

$$PA = LU, \quad Lc = Pb, \quad Ux = c$$

L: unit lower triangular, storing multipliers. U: upper triangular. Without swaps, $P = I$; $P^{-1} = P^T$. Factor once in $O(n^3)$ for a dense $n \times n$ system; each new right-hand side costs $O(n^2)$ to solve.

$$\begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

Watch: Permute b too. Check the factorization by multiplying LU.

04 Invertibility and 2×2 inverse [S04 p2–3; S09 p2–3; S19 p2; S21 p1]

For an $n \times n$ matrix, all are equivalent:

$$A^{-1} \text{ exists} \Leftrightarrow r = n \Leftrightarrow N(A) = \{0\}$$
$$\Leftrightarrow \det A \neq 0 \Leftrightarrow R = I$$

Equivalently: the columns form a basis of $\mathbb{R}^n$; or $Ax = b$ has exactly one solution for **every** b.

$$[A \quad | \quad I] \to [I \quad | \quad A^{-1}]$$

For $ad - bc \neq 0$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Swap the diagonal entries, negate the off-diagonal entries, and divide by the determinant.

05 Subspaces: the closure test [S06 p1–2; S07 p1–2; S12 p1]

A subspace contains zero and is closed under addition and multiplication by **every** real scalar:

$$u, v \in V \Rightarrow \alpha u + \beta v \in V$$

The solutions of $Ax = 0$ form a subspace. A nonempty solution set of $Ax = b \neq 0$ is an affine translate, not a subspace. Intersections of subspaces are subspaces. A union of two distinct lines through zero is not closed under addition.
Watch: Containing zero is necessary, not sufficient.

06 RREF, rank and nullspace [S08 p1–2; S09 p1–3]

Count pivots to obtain r. There are r pivot variables and $n - r$ free variables. Set one free variable to 1 and the others to 0; solve for the pivots. These special solutions form a basis of $N(A)$.

$$\dim N(A) = n - r$$

With pivot columns ordered first:

$$R = \begin{pmatrix} I & F \\ 0 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} -F \\ I \end{pmatrix}, \quad RZ = 0$$

Authored example: if $R = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, let $x_2 = t$; then $x = t(-2, 1, 0)^T$.
Watch: Restore the original variable order after any column permutation.

07 Consistency and solution count [S07 p1; S09 p1–3]

$$Ax = b \text{ solvable} \Leftrightarrow b \in C(A)$$

Equivalently, $\operatorname{rank}(A) = \operatorname{rank}([A \quad | \quad b])$. An augmented row $[0 \ldots 0 \quad | \quad d]$ with $d \neq 0$ means no solution. If consistent, choose a particular solution x_p by setting the free variables to zero; all solutions are

$$x = x_p + z, \quad z \in N(A)$$

Rank case	Solutions for given b
$r = m = n$	exactly 1
$r = n < m$	0 or 1
$r = m < n$	infinitely many
$r < m, r < n$	0 or infinitely many

中文提示：先查是否有解，再查自由变量；相容性取决于 b，唯一性取决于 $N(A)$。

08 Independence, basis, dimension [S10 p1–3]

Independent: $\sum c_j v_j = 0$ implies every $c_j = 0$.
Span: all linear combinations fill the target space.
Basis: independent and spanning. **Dimension:** number of vectors in any basis. A basis of $C(A)$ uses the pivot columns of the **original** A. A basis of the row space uses the nonzero rows of R.
中文提示：消元后的主元位置用于选原矩阵的列；行变换保留行空间和零空间，但通常改变列空间。

09 Four fundamental subspaces [S11 p1–3; S15 p1–2; S09 p1]

Subspace	Ambient	Dim.
$C(A)$	$\mathbb{R}^m$	r
$N(A^T)$	$\mathbb{R}^m$	$m - r$
$C(A^T)$	$\mathbb{R}^n$	r
$N(A)$	$\mathbb{R}^n$	$n - r$

$$C(A)^\perp = N(A^T), \quad C(A^T)^\perp = N(A)$$

The row space and nullspace are orthogonal complements in $\mathbb{R}^n$; the column space and left nullspace are complements in $\mathbb{R}^m$. **Consistency test:** $y^Tb = 0$ for every $y \in N(A^T)$.

10 Matrix spaces and rank one [S11 p3; S12 p1–3]

Matrices themselves can be vectors: $\dim \mathbb{R}^{m\times n} = mn$. Symmetric and upper-triangular $n \times n$ spaces each have dimension $\frac{n(n+1)}{2}$; diagonal matrices have dimension n. Rank one: $A = uv^T$ for nonzero u, v; every column is a multiple of u. A fixed-rank collection is generally not a subspace.

$$\dim(S + T) = \dim S + \dim T - \dim(S \cap T)$$

Count independent degrees of freedom, not the number of displayed entries.

11 Graphs and network matrices [S12 p3; S13 p1–4]

Incidence matrix B: one row per edge, one column per node. Edge $i \to j$: put -1 at i, $+1$ at j. Bx gives potential differences. For a connected graph with v nodes and e edges:

$$\operatorname{rank}(B) = v - 1, \quad N(B) = \operatorname{span}\{\mathbf{1}\}$$
$$\dim N(B^T) = e - v + 1$$

The last quantity counts independent cycles. A tree has $v - 1$ edges and no cycles. Kirchhoff's current law with no injected current: $B^Ty = 0$. With conductance matrix D and injection f:

$$d = Bx, \quad y = Dd, \quad B^TDBx = f$$

Watch: Nodes index columns; edges index rows.

12 Orthogonality and normal equations [S15 p1–2; S17 p3]

$$u \perp v \Leftrightarrow u^Tv = 0, \quad \|u\|^2 = u^Tu$$

Orthogonal subspaces require **every** vector in one to be orthogonal to every vector in the other.

$$N(A^TA) = N(A)$$
$$z^TA^TAz = \|Az\|^2$$

Hence A^TA is invertible exactly when the columns of A are independent. This condition makes the least-squares coefficient vector unique.

13 Orthogonal projections [S16 p1–2; S17 p1]

Onto the line spanned by nonzero a:

$$p = a\frac{a^Tb}{a^Ta}, \quad P = \frac{aa^T}{a^Ta}$$

Onto $C(A)$ when A has independent columns:

$$P = A(A^TA)^{-1}A^T, \quad p = Pb$$
$$P^T = P, \quad P^2 = P, \quad e = (I - P)b$$

$p \in C(A)$ and $e \perp C(A)$. If $b \in C(A)$, then $Pb = b$. If $b \perp C(A)$, then $Pb = 0$. **Authored example:** $a = (1, 2)^T$, $b = (3, 0)^T$ gives $p = \left(\frac{3}{5}, \frac{6}{5}\right)^T$; check $a^{T(b-p)} = 0$.
Watch: Do not use the inverse formula when A is column-rank deficient.

14 Least squares and line fitting [S16 p3; S17 p1–3]

Minimize the residual sum of squares:

$$\min_x \|b - Ax\|^2 \Rightarrow A^TA\hat{x} = A^Tb$$

For full column rank, $\hat{x} = (A^TA)^{-1}A^Tb$. Fit $b_i \approx C + Dt_i$ using design row $[1, t_i]$. With sample means μ_t, μ_b:

$$\hat{D} = \frac{\sum_i (t_i - \mu_t)(b_i - \mu_b)}{\sum_i (t_i - \mu_t)^2}, \quad \hat{C} = \mu_b - \hat{D}\mu_t$$

At least two distinct t_i are needed. Source example: $(t, b) = (1, 1), (2, 2), (3, 2)$ gives $\hat{C} = \frac{2}{3}$, $\hat{D} = \frac{1}{2}$, $\|e\|^2 = \frac{1}{6}$.
中文提示：数据图上最小化竖直残差；投影垂直性是在整个观测向量的 $\mathbb{R}^m$ 空间，不是点到拟合线的距离。

15 Orthonormal columns [S18 p1–3; S15 p1]

$$q_i^Tq_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}, \quad Q^TQ = I$$

Then $P = QQ^T$ and the projection coordinates are $\hat{x} = Q^Tb$. Only for **square** Q do we have $Q^{-1} = Q^T$ and $QQ^T = I$. For the orthogonal split $b = p + e$:

$$p^Te = 0, \quad \|b\|^2 = \|p\|^2 + \|e\|^2$$

Watch: Orthonormal columns do not make a rectangular matrix invertible.

16 Gram–Schmidt and QR [S18 p2–3]

Start with independent columns $a_1, \ldots, a_n$. Remove earlier projections; normalize the remainder:

$$u_j = a_j - \sum_{i<j}(q_i^Ta_j)q_i, \quad q_j = \frac{u_j}{\|u_j\|}$$

$$A = QR, \quad R = Q^TA$$

R is upper triangular: $r_{ij} = q_i^Ta_j$ for $i < j$, $r_{jj} = \|u_j\|$. Solve least squares via $R\hat{x} = Q^Tb$. Source example: $a_1 = (1, 1, 1)^T$, $a_2 = (1, 0, 2)^T$:

$$q_1 = \frac{(1,1,1)^T}{\sqrt{3}}, \quad q_2 = \frac{(0,-1,1)^T}{\sqrt{2}}$$

中文提示：先减去已保留方向的投影，再归一化；不是把原向量分别除以长度。
Watch: $u_j = 0$ signals dependence; never divide by zero.

17 Determinants and elimination [S19 p1–3]

For square matrices: swap two rows → change sign; scale one row by c → scale the determinant by c; add a multiple of another row → no change.

$$\det I = 1, \quad \det(AB) = \det A \det B$$
$$\det A^T = \det A, \quad \det(cA) = c^n \det A$$

Dependent rows or columns give determinant zero. If A is invertible, $\det A^{-1} = \frac{1}{\det A}$. For $PA = LU$ with unit diagonal L and s swaps:

$$\det A = (-1)^s \prod_{i=1}^{n} u_{ii}$$

Watch: Generally $\det(A + B) \neq \det A + \det B$.

18 Cofactors and determinant formulas [S20 p1–3]

Delete row i and column j to form minor M_{ij}:

$$C_{ij} = (-1)^{i+j} \det M_{ij}$$
$$\det A = \sum_{j=1}^{n} a_{ij}C_{ij}$$

Expand along a sparse row/column; signs start $+$ at the upper-left. S_n: all $n!$ permutations.

$$\det A = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) \prod_{i=1}^{n} a_{i,\pi(i)}$$

For tridiagonal all-ones bands: $d_n = d_{n-1} - d_{n-2}$, $d_1 = 1$, $d_2 = 0$ (source example).
Watch: The 3×3 diagonal shortcut does not extend to 4×4.

19 Inverse, Cramer and volume [S21 p1–4]

Let C be the cofactor matrix. When $\det A \neq 0$:

$$A^{-1} = \frac{C^T}{\det A}, \quad x_j = \frac{\det A_j}{\det A}$$

A_j: replace **column** j by b. Elimination is usually more efficient. Column parallelepiped: volume $|\det A|$. Triangle: half the absolute determinant of its two edges. **Authored:** edges $(2, 0)^T, (1, 3)^T$ → parallelogram area 6; triangle area 3.
Watch: Transpose the cofactor matrix. Volume is nonnegative; the sign of the determinant records orientation.

20 Exam workflow and checks [S14 p1–5 + synthesis]

Solve: $[A \quad | \quad b]$ → consistency → $x_p + N(A)$.
Basis: name the space → pivots / special solutions → check dimension and independence. **Fit:** design matrix → normal equations / QR → $A^Te = 0$.
Checks: $Ax = b$; $Az = 0$; $Q^TQ = I$; $P^2 = P$; determinant sign after swaps. **Answer cue:** State the rank condition before using an inverse.
Watch: Consistent ≠ unique. Full column rank need not fill $\mathbb{R}^m$.